

Electrostatic Potential and Capacitance

Chapter 2 · All Questions and Solutions



Part 1 · Examples

Q

1

Potential and work done near a point charge



Does the answer depend on the path along which the charge is brought?

Part (a)

Calculate the potential at a point P due to a charge of 4×10^{-7} C located 9 cm away.

Key formula

$$V = \frac{1}{4\pi\epsilon_0} \frac{Q}{r} = 9 \times 10^9 \text{ Nm}^2 \text{ C}^{-2} \times \frac{4 \times 10^{-7} \text{ C}}{0.09 \text{ m}} = 4 \times 10^4 \text{ V}$$

$$V = 4 \times 10^4 \text{ V} //$$

Part (b)

Hence obtain the work done in bringing a charge of 2×10^{-9} C from infinity to the point P. Does the answer depend on the path along which the charge is brought?

Key formula

$$W = qV = 2 \times 10^{-9} \text{ C} \times 4 \times 10^4 \text{ V} = 8 \times 10^{-5} \text{ J}$$

Conclusion

No, the work done does not depend on the path.

Any small path can be split into two perpendicular parts: one along r , and one perpendicular to r .

The work done along the perpendicular part is zero.

$$W = 8 \times 10^{-5} \text{ J}; \text{ no, work done is path independent} //$$

Q

2

Point of zero potential between two charges



Two charges 3×10^{-8} C and -2×10^{-8} C are located 15 cm apart. At what point on the line joining the two charges is the electric potential zero? Take the potential at infinity to be zero.

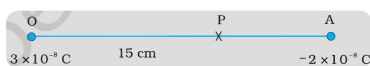


FIGURE 2.7

Given

Let us take the origin O at the location of the positive charge, with the line joining the two charges as the x-axis and the negative charge on the right of the origin (Fig. 2.7).

Let P be the required point on the x-axis where the potential is zero, with x its x-coordinate.

The potentials from the two charges cannot add up to zero for $x < 0$, so x must be positive.

Substitute

If x lies between O and A, we have

$$\frac{1}{4\pi\epsilon_0} \frac{3 \times 10^{-8}}{x \times 10^{-2}} - \frac{2 \times 10^{-8}}{(15-x) \times 10^{-2}} = 0$$

where x is in cm.

That is,

$$\frac{3}{x} - \frac{2}{15-x} = 0$$

which gives x = 9 cm.

If x lies on the extended line OA, the required condition is

$$\frac{3}{x} - \frac{2}{x-15} = 0$$

which gives

$$x = 45 \text{ cm}$$

Conclusion

So the electric potential is zero at 9 cm and at 45 cm from the positive charge, on the side of the negative charge.

This uses the convention that potential is zero at infinity.

zero at 9 cm and at 45 cm from the positive charge, on the side of the negative charge

///

Q

3

Signs of
potential and
work along
field lines

(())

Figures 2.8

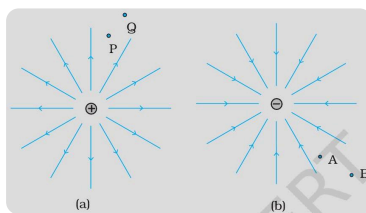


FIGURE 2.8

Part (a)

Give the signs of the potential difference $V_P - V_Q$; $V_B - V_A$.

Conclusion

$$\text{As } V \propto \frac{1}{r}, V_P > V_Q.$$

Thus, $(V_P - V_Q)$ is positive.

Also V_B is less negative than V_A .

Thus, $V_B > V_A$ or $(V_B - V_A)$ is positive.

both $(V_P - V_Q)$ and $(V_B - V_A)$ are positive

///

Part (b)

Give the sign of the potential energy difference of a small negative charge between the points Q and P ; A and B .

Conclusion

A small negative charge will be attracted towards positive charge.

The negative charge moves from higher potential energy to lower potential energy.

Therefore the sign of potential energy difference of a small negative charge between Q and P is positive.

Similarly, $(P.E.)_A > (P.E.)_B$ and hence sign of potential energy differences is positive.

positive in both cases ///

Part (c)

Give the sign of the work done by the field in moving a small positive charge from Q to P .

Conclusion

In moving a small positive charge from Q to P , work has to be done by an external agency against the electric field.

Therefore, work done by the field is negative.

negative ///

Part (d)

Give the sign of the work done by the external agency in moving a small negative charge from B to A.

Conclusion

In moving a small negative charge from B to A work has to be done by the external agency.

It is positive.

positive ///

Part (e)

Does the kinetic energy of a small negative charge increase or decrease in going from B to A?

Conclusion

Due to force of repulsion on the negative charge, velocity decreases and hence the kinetic energy decreases in going from B to A.

decreases ///

Q

4

Work to assemble charges on a square



Four charges are arranged at the corners of a square ABCD of side d, as shown in Fig. 2.15. How much extra work is needed to do this?

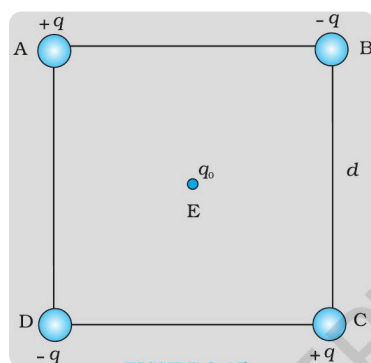


FIGURE 2.15

Part (a)

Find the work required to put together this arrangement.

Given

The work done depends only on the final arrangement of the charges, not on how we bring them together.

So we find the work needed for one way of assembling them: bring +q to A first, then the charges -q, +q and -q to B, C and D respectively.

We add up the work in four steps:

Substitute

Bringing charge $+q$ to A, with no charge anywhere else, needs zero work.

Bringing $-q$ to B, with $+q$ already at A, needs (charge at B) $\times$ (potential at B due to charge $+q$ at A):

$$= -q \times \frac{q}{4\pi\epsilon_0 d} = -\frac{q^2}{4\pi\epsilon_0 d}$$

Bringing charge $+q$ to C, with $+q$ at A and $-q$ at B, needs (charge at C) $\times$ (potential at C due to the charges at A and B):

$$= +q \frac{+q}{4\pi\epsilon_0 d \sqrt{2}} + \frac{-q}{4\pi\epsilon_0 d}$$

$$= \frac{-q^2}{4\pi\epsilon_0 d} 1 - \frac{1}{\sqrt{2}}$$

Bringing $-q$ to D, with $+q$ at A, $-q$ at B, and $+q$ at C, needs (charge at D) $\times$ (potential at D due to the charges at A, B and C):

$$= -q \frac{+q}{4\pi\epsilon_0 d} + \frac{-q}{4\pi\epsilon_0 d \sqrt{2}} + \frac{q}{4\pi\epsilon_0 d}$$

$$= \frac{-q^2}{4\pi\epsilon_0 d} 2 - \frac{1}{\sqrt{2}}$$

Conclusion

Adding the work from steps (i)-(iv) gives the total work required:

$$= \frac{-q^2}{4\pi\epsilon_0 d} (0) + (1) + 1 - \frac{1}{\sqrt{2}} + 2 - \frac{1}{\sqrt{2}}$$

$$= \frac{-q^2}{4\pi\epsilon_0 d} (4 - \sqrt{2})$$

This depends only on how the charges are arranged, not on the order in which we bring them together — by definition, this is the total electrostatic energy of the charges.

(Students may try calculating the same energy in a different order and check that it comes out the same.)

$$\frac{-q^2}{4\pi\epsilon_0 d} (4 - \sqrt{2}) //$$

Part (b)

A charge q_0 is brought to the centre E of the square, the four charges being held fixed at its corners. How much extra work is needed to do this?

Conclusion

The extra work necessary to bring a charge q_0 to the point E when the four charges are at A, B, C and D is $q_0 \times$ (electrostatic potential at E due to the charges at A, B, C and D).

The electrostatic potential at E is clearly zero since potential due to A and C is cancelled by that due to B and D.

Hence, no work is required to bring any charge to point E.

$$\text{zero} //$$

Q

5

Potential energy of two charges in a field



What would the electrostatic energy of the configuration be?

Part (a)

Determine the electrostatic potential energy of a system consisting of two charges $7 \mu\text{C}$ and $-2 \mu\text{C}$ (and with no external field) placed at $(-9 \text{ cm}, 0, 0)$ and $(9 \text{ cm}, 0, 0)$ respectively.

Conclusion

$$U = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r} = 9 \times 10^9 \times \frac{7 \times (-2) \times 10^{-12}}{0.18} = -0.7 \text{ J.}$$

$$U = -0.7 \text{ J} //$$

Part (b)

How much work is required to separate the two charges infinitely away from each other?

Conclusion

$$W = U_2 - U_1 = 0 - U = 0 - (-0.7) = 0.7 \text{ J.}$$

$$W = 0.7 \text{ J} //$$

Part (c)

Suppose that the same system of charges is now placed in an external electric field $E = A(1/r^2)$; $A = 9 \times 10^5 \text{ NC}^{-1} \text{ m}^2$. What would the electrostatic energy of the configuration be?

Conclusion

The mutual interaction energy of the two charges remains unchanged.

In addition, there is the energy of interaction of the two charges with the external electric field.

We find,

$$q_1 V(\mathbf{r}_1) + q_2 V(\mathbf{r}_2) = A \frac{7 \mu\text{C}}{0.09 \text{ m}} + A \frac{-2 \mu\text{C}}{0.09 \text{ m}}$$

and the net electrostatic energy is

$$q_1 V(\mathbf{r}_1) + q_2 V(\mathbf{r}_2) + \frac{q_1 q_2}{4\pi\epsilon_0 r_{12}} = A \frac{7 \mu\text{C}}{0.09 \text{ m}} + A \frac{-2 \mu\text{C}}{0.09 \text{ m}} - 0.7 \text{ J}$$

$$= 70 - 20 - 0.7 = 49.3 \text{ J}$$

$$49.3 \text{ J} //$$

Q

6

Heat released reorienting dipoles in a field



A molecule of a substance has a permanent electric dipole moment of magnitude 10^{-29} C m . A mole of this substance is polarised (at low temperature) by applying a strong electrostatic field of magnitude 10^6 V m^{-1} . The direction of the field is suddenly changed by an angle of 60° . Estimate the heat released by the substance in aligning its dipoles along the new direction of the field. For simplicity, assume 100% polarisation of the sample.

Given

The dipole moment of each molecule is $= 10^{-29} \text{ C m}$.

One mole of the substance has 6×10^{23} molecules, so the total dipole moment of all the molecules is $p = 6 \times 10^{23} \times 10^{-29} \text{ C m} = 6 \times 10^{-6} \text{ C m}$.

Substitute

Initial potential energy, $U_i = -p E \cos \theta = -6 \times 10^{-6} \times 10^6 \cos 0^\circ = -6 \text{ J}$.

Final potential energy (when $\theta = 60^\circ$), $U_f = -6 \times 10^{-6} \times 10^6 \cos 60^\circ = -3 \text{ J}$.

Change in potential energy = $-3 \text{ J} - (-6 \text{ J}) = 3 \text{ J}$.

Conclusion

So there is a loss in potential energy — this must be the energy released by the substance as heat while its dipoles align.

3 J //

Q

7

**Conductors,
insulators
and
electrostatic
safety**

(())

A man standing on the ground touches the same line and gets a fatal shock. Why?

Part (a)

A comb run through one's dry hair attracts small bits of paper. Why? What happens if the hair is wet or if it is a rainy day? (Remember, a paper does not conduct electricity.)

Conclusion

This is because the comb gets charged by friction.

The molecules in the paper gets polarised by the charged comb, resulting in a net force of attraction.

If the hair is wet, or if it is rainy day, friction between hair and the comb reduces.

The comb does not get charged and thus it will not attract small bits of paper.

charging by friction polarises the paper's molecules, giving a net attractive force; wet hair reduces friction, so the comb no longer charges and no longer attracts the paper

//

Part (b)

Ordinary rubber is an insulator. But special rubber tyres of aircraft are made slightly conducting. Why is this necessary?

Conclusion

To let the charge built up by friction flow safely to the ground.

If too much static electricity builds up, it can spark and start a fire.

to conduct away friction-produced charge before it sparks and causes fire

//

Part (c)

Vehicles carrying inflammable materials usually have metallic ropes touching the ground during motion. Why?

Conclusion

Reason similar to (b).

same reason as (b) - to conduct away friction-produced charge safely to the ground

//

Part (d)

A bird perches on a bare high power line, and nothing happens to the bird. A man standing on the ground touches the same line and gets a fatal shock. Why?

Conclusion

Current passes only when there is difference in potential.

the bird's two feet are at (nearly) the same potential, so no current flows through it; the man, in contact with the line and the ground, spans a large potential difference

///

Q

8

Capacitance with a partial dielectric slab



A slab of material of dielectric constant K has the same area as the plates of a parallel-plate capacitor but has a thickness $\frac{3}{4}d$, where d is the separation of the plates. How is the capacitance changed when the slab is inserted between the plates?

Given

Let $E_0 = \frac{V_0}{d}$ be the electric field between the plates when there is no dielectric and the potential difference is V_0 .

If the dielectric is now inserted, the electric field in the dielectric will be $E = \frac{E_0}{K}$.

The potential difference will then be

$$V = E_0 \left(\frac{1}{4}d \right) + \frac{E_0}{K} \left(\frac{3}{4}d \right)$$

$$= E_0 d \left(\frac{1}{4} + \frac{3}{4K} \right) = V_0 \frac{K+3}{4K}$$

The potential difference decreases by the factor $\frac{(K+3)}{4K}$ while the free charge Q_0 on the plates remains unchanged.

The capacitance thus increases

Key formula

$$C = \frac{Q_0}{V} = \frac{4K}{K+3} \frac{Q_0}{V_0} = \frac{4K}{K+3} C_0$$

$$C = \frac{4K}{K+3} C_0 \quad ///$$

Q

9

Equivalent capacitance and charge in a network



A network of four $10 \mu\text{F}$ capacitors is connected to a 500 V supply, as shown in Fig. 2.29. Determine (a) the equivalent capacitance of the network and (b) the charge on each capacitor. (Note, the charge on a capacitor is the charge on the plate with higher potential, equal and opposite to the charge on the plate with lower potential.)

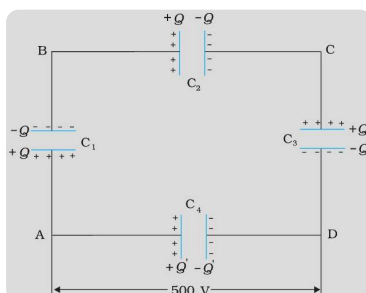


FIGURE 2.29

Part (a)

Determine the equivalent capacitance of the network.

Given

In the given network, C_1 , C_2 and C_3 are connected in series.

The effective capacitance C' of these three capacitors is given by

Key formula

$$\frac{1}{C'} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}$$

Conclusion

For $C_1 = C_2 = C_3 = 10 \mu\text{F}$, $C' = \frac{10}{3} \mu\text{F}$.

The network has C' and C_4 connected in parallel.

Thus, the equivalent capacitance C of the network is

$$C = C' + C_4 = \frac{10}{3} + 10 \mu\text{F} = 13.3 \mu\text{F}$$

$$\boxed{C = 13.3 \mu\text{F}} //$$

Part (b)

Determine the charge on each capacitor.

Given

Clearly, from the figure, the charge on each of the capacitors, C_1 , C_2 and C_3 is the same, say Q .

Let the charge on C_4 be Q' .

Now, since the potential difference across AB is $\frac{Q}{C_1}$, across BC is $\frac{Q}{C_2}$, across

CD is $\frac{Q}{C_3}$, we have

$$\frac{Q}{C_1} + \frac{Q}{C_2} + \frac{Q}{C_3} = 500 \text{ V}$$

$$\text{Also, } \frac{Q'}{C_4} = 500 \text{ V.}$$

Conclusion

This gives for the given value of the capacitances,

$$Q = 500 \text{ V} \times \frac{10}{3} \mu\text{F} = 1.7 \times 10^{-3} \text{ C and}$$

$$Q' = 500 \text{ V} \times 10 \mu\text{F} = 5.0 \times 10^{-3} \text{ C}$$

$$\boxed{Q = 1.7 \times 10^{-3} \text{ C on } C_1, C_2, C_3; Q' = 5.0 \times 10^{-3} \text{ C on } C_4} //$$

Q

10

Energy
stored when
capacitors
are
connected



What is the electrostatic energy stored by the system?

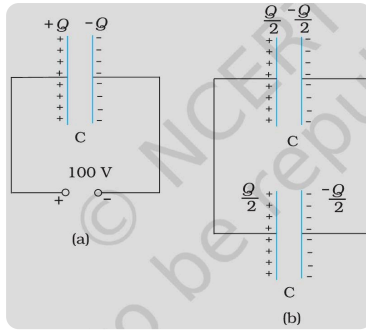


FIGURE 2.31

Part (a)

A 900 pF capacitor is charged by 100 V battery [Fig. 2.31(a)]. How much electrostatic energy is stored by the capacitor?

Conclusion

The charge on the capacitor is

$$Q = C V = 900 \times 10^{-12} \text{ F} \times 100 \text{ V} = 9 \times 10^{-8} \text{ C}$$

The energy stored by the capacitor is

$$\begin{aligned} &= \frac{1}{2} C V^2 = \frac{1}{2} Q V \\ &= \frac{1}{2} \times 9 \times 10^{-8} \text{ C} \times 100 \text{ V} = 4.5 \times 10^{-6} \text{ J} \end{aligned}$$

$$4.5 \times 10^{-6} \text{ J} //$$

Part (b)

The capacitor is disconnected from the battery and connected to another 900 pF capacitor [Fig. 2.31(b)]. What is the electrostatic energy stored by the system?

Conclusion

In the steady state, both capacitors have their positive plates at the same potential, and their negative plates at the same potential.

Let this common potential difference be V' , so the charge on each capacitor is $Q' = C V'$.

Charge conservation gives $Q' = \frac{Q}{2}$, so $V' = \frac{V}{2}$.

The total energy of the system is

$$= 2 \times \frac{1}{2} Q' V' = \frac{1}{4} Q V = 2.25 \times 10^{-6} \text{ J}$$

So no charge is lost in going from (a) to (b), but the final energy is only half the initial energy.

$2.25 \times 10^{-6} \text{ J}$ (half the initial energy; the rest is lost as heat and electromagnetic radiation during the transient)

///

Where did the energy go?

There is a brief transient period before the system settles into (b).

During this time, a transient current flows from the first capacitor to the second, and energy is lost as heat and electromagnetic radiation.

Q

1

Zero-potential point between two charges

(())

Two charges $5 \times 10^{-8} \text{ C}$ and $-3 \times 10^{-8} \text{ C}$ are located 16 cm apart. At what point(s) on the line joining the two charges is the electric potential zero? Take the potential at infinity to be zero.

Given

There are two charges,

$$q_1 = 5 \times 10^{-8} \text{ C}$$

$$q_2 = -3 \times 10^{-8} \text{ C}$$

Distance between the two charges, $d = 16 \text{ cm} = 0.16 \text{ m}$

Let the electric potential (V) at point P be zero.

Potential at point P is the sum of potentials caused by charges q_1 and q_2 respectively.

$$\therefore V = \frac{q_1}{4\pi\epsilon_0 r} + \frac{q_2}{4\pi\epsilon_0(d-r)}$$

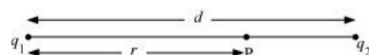
Where,

ϵ_0 = Permittivity of free space

Substitute

Consider a point P on the line joining the two charges, as shown in the given figure.

r = Distance of point P from charge q_1 .



For $V = 0$, equation (i) reduces to

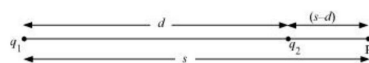
$$\begin{aligned} \frac{q_1}{4\pi\epsilon_0 r} &= - \frac{q_2}{4\pi\epsilon_0(d-r)} \\ \frac{q_1}{r} &= \frac{-q_2}{d-r} \\ \frac{5 \times 10^{-8}}{r} &= \frac{(-3 \times 10^{-8})}{(0.16-r)} \\ \frac{0.16}{r} - 1 &= \frac{3}{5} \\ \frac{0.16}{r} &= \frac{8}{5} \end{aligned}$$

$$\therefore r = 0.1 \text{ m} = 10 \text{ cm}$$

Therefore, the potential is zero at a distance of 10 cm from the positive charge between the charges.

Conclusion

Suppose point P is outside the system of two charges at a distance s from the negative charge, where potential is zero, as shown in the following figure.



For this arrangement, potential is given by,

$$V = \frac{q_1}{4\pi\epsilon_0 s} + \frac{q_2}{4\pi\epsilon_0(s-d)}$$

For $V = 0$, equation (ii) reduces to

$$\frac{q_1}{4\pi\epsilon_0 s} = -\frac{q_2}{4\pi\epsilon_0 (s-d)}$$

$$\frac{q_1}{s} = \frac{-q_2}{s-d}$$

$$\frac{5 \times 10^{-8}}{s} = -\frac{(-3 \times 10^{-8})}{(s-0.16)}$$

$$1 - \frac{0.16}{s} = \frac{3}{5}$$

$$\frac{0.16}{s} = \frac{2}{5}$$

$$\therefore s = 0.4 \text{ m} = 40 \text{ cm}$$

Therefore, the potential is zero at a distance of 40 cm from the positive charge outside the system of charges.

zero at 10 cm from the positive charge (between the charges) and at 40 cm from the positive charge (outside the system, beyond the negative charge)

///

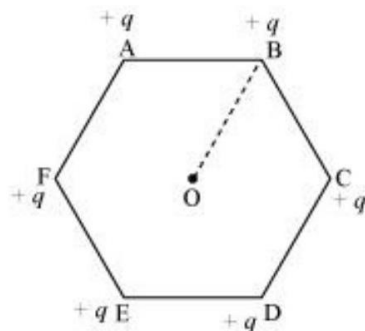
Q

2

Potential at the centre of a charged hexagon



A regular hexagon of side 10 cm has a charge $5 \mu\text{C}$ at each of its vertices. Calculate the potential at the centre of the hexagon.



Given

The given figure shows six equal amount of charges, q , at the vertices of a regular hexagon.

Where,

Charge, $q = 5 \mu\text{C} = 5 \times 10^{-6} \text{ C}$

Side of the hexagon, $l = AB = BC = CD = DE = EF = FA = 10 \text{ cm}$

Distance of each vertex from centre O, $d = 10 \text{ cm}$

Key formula

Electric potential at point O,

$$V = \frac{6 \times q}{4\pi\epsilon_0 d}$$

Substitute

Where,

ϵ_0 = Permittivity of free space

$$\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ N C}^{-2} \text{ m}^{-2}$$

$$\therefore V = \frac{6 \times 9 \times 10^9 \times 5 \times 10^{-6}}{0.1}$$

$$= 2.7 \times 10^6 \text{ V}$$

Conclusion

Therefore, the potential at the centre of the hexagon is $2.7 \times 10^6 \text{ V}$.

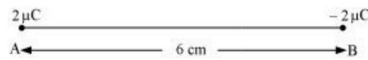
$$2.7 \times 10^6 \text{ V} //$$

Q**3**

Equipotential surface of two equal opposite charges



Two charges $2 \mu\text{C}$ and $-2 \mu\text{C}$ are placed at points A and B 6 cm apart.



Part (a)

Identify an equipotential surface of the system.

Conclusion

An equipotential surface is the plane on which total potential is zero everywhere.

This plane is normal to line AB.

The plane is located at the mid-point of line AB because the magnitude of charges is the same.

the plane normal to AB through its mid-point //

Part (b)

What is the direction of the electric field at every point on this surface?

Conclusion

The direction of the electric field at every point on this surface is normal to the plane in the direction of AB.

normal to the plane, in the direction of AB //

Q**4**

Electric field of a charged spherical conductor



A spherical conductor of radius 12 cm has a charge of $1.6 \times 10^{-7} \text{ C}$ distributed uniformly on its surface. What is the electric field

Part (a)

inside the sphere

Conclusion

Radius of the spherical conductor, $r = 12 \text{ cm} = 0.12 \text{ m}$

Charge is uniformly distributed over the conductor, $q = 1.6 \times 10^{-7} \text{ C}$

Electric field inside a spherical conductor is zero.

This is because if there is field inside the conductor, then charges will move to neutralize it.

zero //

Part (b)

just outside the sphere

Key formula

Electric field E just outside the conductor is given by the relation,

$$E = \frac{q}{4\pi\epsilon_0 r^2}$$

Conclusion

Where,

ϵ_0 = Permittivity of free space

$$\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$$
$$\therefore E = \frac{1.6 \times 10^{-7} \times 9 \times 10^9}{(0.12)^2}$$
$$= 10^5 \text{ N C}^{-1}$$

Therefore, the electric field just outside the sphere is 10^5 N C^{-1} .

$$\boxed{10^5 \text{ N C}^{-1}} //$$

Part (c)

at a point 18 cm from the centre of the sphere?

Given

Electric field at a point 18 m from the centre of the sphere = E_1

Distance of the point from the centre, $d = 18 \text{ cm} = 0.18 \text{ m}$

Key formula

$$E_1 = \frac{q}{4\pi\epsilon_0 d^2}$$
$$= \frac{9 \times 10^9 \times 1.6 \times 10^{-7}}{(18 \times 10^{-2})^2}$$
$$= 4.4 \times 10^4 \text{ N / C}$$

Conclusion

Therefore, the electric field at a point 18 cm from the centre of the sphere is

$$4.4 \times 10^4 \text{ N / C}.$$

$$\boxed{4.4 \times 10^4 \text{ N / C}} //$$

Q

5

Capacitance
with halved
gap and
dielectric

(())

A parallel plate capacitor with air between the plates has a capacitance of 8 pF ($1 \text{ pF} = 10^{-12} \text{ F}$). What will be the capacitance if the distance between the plates is reduced by half, and the space between them is filled with a substance of dielectric constant 6?

Given

Capacitance between the parallel plates of the capacitor, $C = 8 \text{ pF}$

Initially, distance between the parallel plates was d and it was filled with air.

Dielectric constant of air, $k = 1$

Capacitance, C , is given by the formula,

Key formula

$$C = \frac{k\epsilon_0 A}{d}$$
$$= \frac{\epsilon_0 A}{d}$$

Substitute

Where,

A = Area of each plate

ϵ_0 = Permittivity of free space

If distance between the plates is reduced to half, then new distance, $d = \frac{d}{2}$

Dielectric constant of the substance filled in between the plates, $k' = 6$

Hence, capacitance of the capacitor becomes

Key formula

$$C' = \frac{k' \epsilon_0 A}{d'} = \frac{6 \epsilon_0 A}{\frac{d}{2}}$$

Conclusion

Taking ratios of equations (i) and (ii), we obtain

$$\begin{aligned} C' &= 2 \times 6 C \\ &= 12 C \\ &= 12 \times 8 = 96 \text{ pF} \end{aligned}$$

Therefore, the capacitance between the plates is 96 pF.

96 pF ///

Q

6

Series capacitors:
total capacitance and voltage



Three capacitors each of capacitance 9 pF are connected in series.

Part (a)

What is the total capacitance of the combination?

Conclusion

Capacitance of each of the three capacitors, $C = 9 \text{ pF}$

Equivalent capacitance (C') of the combination of the capacitors is given by the relation,

$$\begin{aligned} \frac{1}{C'} &= \frac{1}{C} + \frac{1}{C} + \frac{1}{C} \\ &= \frac{1}{9} + \frac{1}{9} + \frac{1}{9} = \frac{3}{9} = \frac{1}{3} \\ \therefore C' &= 3 \mu\text{F} \end{aligned}$$

Therefore, total capacitance of the combination is $3 \mu\text{F}$.

3 μF ///

Part (b)

What is the potential difference across each capacitor if the combination is connected to a 120 V supply?

Conclusion

Supply voltage, $V = 120 \text{ V}$

Potential difference (V') across each capacitor is equal to one-third of the supply voltage.

$$\therefore V' = \frac{V}{3} = \frac{120}{3} = 40 \text{ V}$$

Therefore, the potential difference across each capacitor is 40 V.

40 V ///

Q

7

Parallel capacitors:
total
capacitance
and charge



Three capacitors of capacitances 2 pF, 3 pF and 4 pF are connected in parallel.

Part (a)

What is the total capacitance of the combination?

Conclusion

Capacitances of the given capacitors are

$$C_1 = 2 \text{ pF}$$

$$C_2 = 3 \text{ pF}$$

$$C_3 = 4 \text{ pF}$$

For the parallel combination of the capacitors, equivalent capacitor C' is given by the algebraic sum,

$$C' = 2+3+4 = 9 \text{ pF}$$

Therefore, total capacitance of the combination is 9 pF.

$$\boxed{9 \text{ pF}} //$$

Part (b)

Determine the charge on each capacitor if the combination is connected to a 100 V supply.

Conclusion

Supply voltage, $V = 100 \text{ V}$

The voltage through all the three capacitors is same $= V = 100 \text{ V}$

Charge on a capacitor of capacitance C and potential difference V is given by the relation,

$$q = V C$$

For $C = 2 \text{ pF}$,

$$\text{Charge} = V C = 100 \times 2 = 200 \text{ pC} = 2 \times 10^{-10} \text{ C}$$

For $C = 3 \text{ pF}$,

$$\text{Charge} = V C = 100 \times 3 = 300 \text{ pC} = 3 \times 10^{-10} \text{ C}$$

For $C = 4 \text{ pF}$,

$$\text{Charge} = V C = 100 \times 4 = 400 \text{ pC} = 4 \times 10^{-10} \text{ C}$$

$$\boxed{2 \times 10^{-10} \text{ C on the 2 pF capacitor; } 3 \times 10^{-10} \text{ C on the 3 pF capacitor; } 4 \times 10^{-10} \text{ C on the 4 pF capacitor}}$$

//

Q

8

Parallel plate capacitor:
capacitance
and charge



In a parallel plate capacitor with air between the plates, each plate has an area of $6 \times 10^{-3} \text{ m}^2$ and the distance between the plates is 3 mm. Calculate the capacitance of the capacitor. If this capacitor is connected to a 100 V supply, what is the charge on each plate of the capacitor?

Given

Area of each plate of the parallel plate capacitor, $A = 6 \times 10^{-3} \text{ m}^2$

Distance between the plates, $d = 3 \text{ mm} = 3 \times 10^{-3} \text{ m}$

Supply voltage, $V = 100 \text{ V}$

Key formula

Capacitance C of a parallel plate capacitor is given by,



$$C = \frac{\epsilon_0 A}{d}$$

Conclusion

Where,

$$\begin{aligned}\epsilon_0 &= \text{Permittivity of free space} \\ &= 8.854 \times 10^{-12} \text{ N}^{-1} \text{ m}^{-2} \text{ C}^{-2}\end{aligned}$$

$$\begin{aligned}\therefore C &= \frac{8.854 \times 10^{-12} \times 6 \times 10^{-3}}{3 \times 10^{-3}} \\ &= 17.71 \times 10^{-12} \text{ F} \\ &= 17.71 \text{ pF}\end{aligned}$$

Potential V is related with the charge q and capacitance C as

$$\begin{aligned}V &= \frac{q}{C} \\ \therefore q &= V C \\ &= 100 \times 17.71 \times 10^{-12} \\ &= 1.771 \times 10^{-9} \text{ C}\end{aligned}$$

Therefore, capacitance of the capacitor is 17.71 pF and charge on each plate is $1.771 \times 10^{-9} \text{ C}$.

$$C = 17.71 \text{ pF}; q = 1.771 \times 10^{-9} \text{ C} //$$

Q

9

Mica sheet
inserted with
supply
connected/disconnected

(())

Explain what would happen if in the capacitor given in Exercise 2.8, a 3 mm thick mica sheet (of dielectric constant = 6) were inserted between the plates,

Part (a)

while the voltage supply remained connected.

Conclusion

Dielectric constant of the mica sheet, $k = 6$

$$\text{Initial capacitance, } C = 1.771 \times 10^{-11} \text{ F}$$

$$\text{New capacitance, } C' = k C = 6 \times 1.771 \times 10^{-11} = 106 \text{ pF}$$

$$\text{Supply voltage, } V = 100 \text{ V}$$

$$\text{New charge, } q' = C' V = 6 \times 1.771 \times 10^{-9} = 1.06 \times 10^{-8} \text{ C}$$

Potential across the plates remains 100 V.

$$C' = 106 \text{ pF}; q' = 1.06 \times 10^{-8} \text{ C}; \text{ potential remains } 100 \text{ V} //$$

Part (b)

after the supply was disconnected.

Conclusion

Dielectric constant, $k = 6$

$$\text{Initial capacitance, } C = 1.771 \times 10^{-11} \text{ F}$$

$$\text{New capacitance, } C' = k C = 6 \times 1.771 \times 10^{-11} = 106 \text{ pF}$$

If supply voltage is removed, then there will be no effect on the amount of charge in the plates.

$$\text{Charge} = 1.771 \times 10^{-9} \text{ C}$$

Potential across the plates is given by,

$$\begin{aligned}\therefore V' &= \frac{q}{C'} \\ &= \frac{1.771 \times 10^{-9}}{106 \times 10^{-12}} \\ &= 16.7 \text{ V}\end{aligned}$$

charge unchanged at $1.771 \times 10^{-9} \text{ C}$; $V' = 16.7 \text{ V}$ //

Q

10

Energy stored in a charged capacitor



A 12pF capacitor is connected to a 50V battery. How much electrostatic energy is stored in the capacitor?

Given

Capacitor of the capacitance, $C = 12 \text{ pF} = 12 \times 10^{-12} \text{ F}$

Potential difference, $V = 50 \text{ V}$

Key formula

Electrostatic energy stored in the capacitor is given by the relation,

$$\begin{aligned}E &= \frac{1}{2} C V^2 \\ &= \frac{1}{2} \times 12 \times 10^{-12} \times (50)^2 \\ &= 1.5 \times 10^{-8} \text{ J}\end{aligned}$$

Conclusion

Therefore, the electrostatic energy stored in the capacitor is $1.5 \times 10^{-8} \text{ J}$.

$1.5 \times 10^{-8} \text{ J}$ //

Q

11

Energy lost connecting two capacitors



A 600pF capacitor is charged by a 200V supply. It is then disconnected from the supply and is connected to another uncharged 600 pF capacitor. How much electrostatic energy is lost in the process?

Given

Capacitance of the capacitor, $C = 600 \text{ pF}$

Potential difference, $V = 200 \text{ V}$

Electrostatic energy stored in the capacitor is given by,

Key formula

$$\begin{aligned}E &= \frac{1}{2} C V^2 \\ &= \frac{1}{2} \times (600 \times 10^{-12}) \times (200)^2 \\ &= 1.2 \times 10^{-5} \text{ J}\end{aligned}$$

Conclusion

If supply is disconnected from the capacitor and another capacitor of capacitance $C = 600 \text{ pF}$ is connected to it, then equivalent capacitance (C') of the combination is given by,

$$\begin{aligned}\frac{1}{C'} &= \frac{1}{C} + \frac{1}{C} \\ &= \frac{1}{600} + \frac{1}{600} = \frac{2}{600} = \frac{1}{300} \\ \therefore C' &= 300 \text{ pF}\end{aligned}$$

New electrostatic energy can be calculated as

$$\begin{aligned}E' &= \frac{1}{2} \times C' \times V^2 \\&= \frac{1}{2} \times 300 \times (200)^2 \\&= 0.6 \times 10^{-5} \text{ J}\end{aligned}$$

Loss in electrostatic energy = $E - E'$

$$\begin{aligned}&= 1.2 \times 10^{-5} - 0.6 \times 10^{-5} \\&= 0.6 \times 10^{-5} \\&= 6 \times 10^{-6} \text{ J}\end{aligned}$$

Therefore, the electrostatic energy lost in the process is $6 \times 10^{-6} \text{ J}$.

$$\boxed{6 \times 10^{-6} \text{ J}} //$$